

Program 1 – Factorial Using math Module

Answer / Python Program

```
import math

num = 5
result = math.factorial(num)
print("The factorial of", num, "is", result)
```

Output

```
The factorial of 5 is 120
```

Note: The programs were formatted as working Python programs. The triangle program uses the complete area formula $\frac{1}{2} \times \text{base} \times \text{height}$, and the maximum-number program avoids recursive shadowing of Python's built-in `max()`.